

Fall 2026 Semester
Cryptographic Application



Lab Environment Basics

LIBERTAS
JUSTITIA
VERITAS

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0. How the Lab Environment Works

- The course repository is opened in VS Code. Docker runs the lab environment.
- A mounted folder connects the two.

```
Your computer
├─ VS Code: read README/PDF and edit code
├─ Docker Desktop
│   └─ Course container
│       └─ /workspace ← course repository is mounted here
```

Key idea: You edit the file once in VS Code. The container sees the saved file through /workspace.



1. What is Docker?

- Docker is a tool for running software in a prepared environment.
- In this course, **Docker provides the same Ubuntu-based lab environment to every student**, even if their own computer is macOS or Windows.

Why we use Docker

- a. Same tools and versions for everyone
- b. Less setup on each student computer
- c. Easier grading and troubleshooting

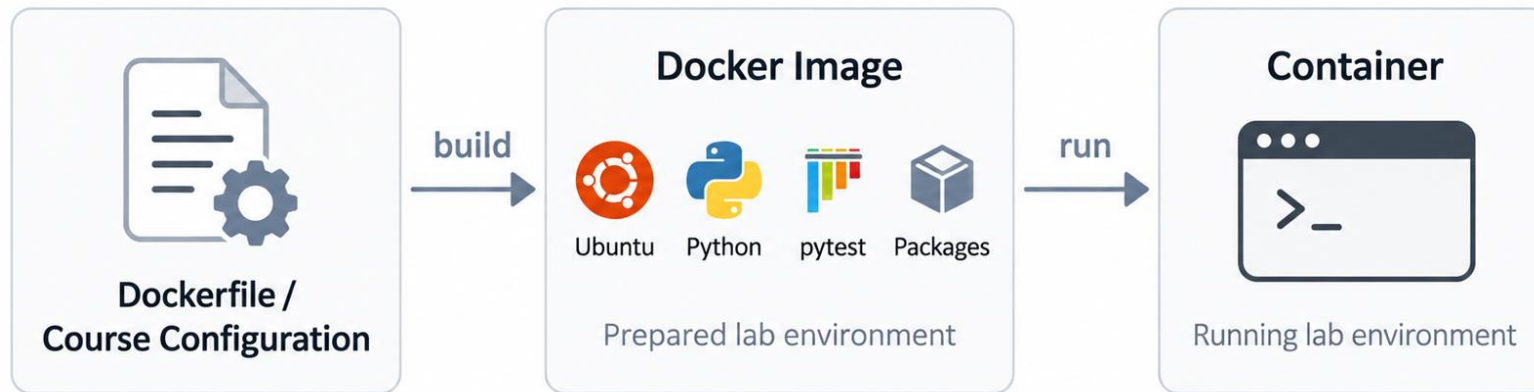
What Docker does not mean

You do not write “Docker code.”
You still write normal Python files for the lab.



2. What is a Container?

- A **container** is a running lab environment created by Docker.
- The course container contains Ubuntu and the tools needed for the labs.
- When you enter the container, your terminal commands run inside this prepared environment.



3. What is a Mount?

- A **mount** makes a folder from your computer visible inside the container.

```
Course repository   on HOST
    ↓ mount
/workspace         inside CONTAINER
```

Because of the mount, the container can run the files that you edited and saved in VS Code.



Image Build vs. Mount

- A **Docker image** contains the prepared environment required for the lab.

```
docker compose -f docker/compose.yml build
```

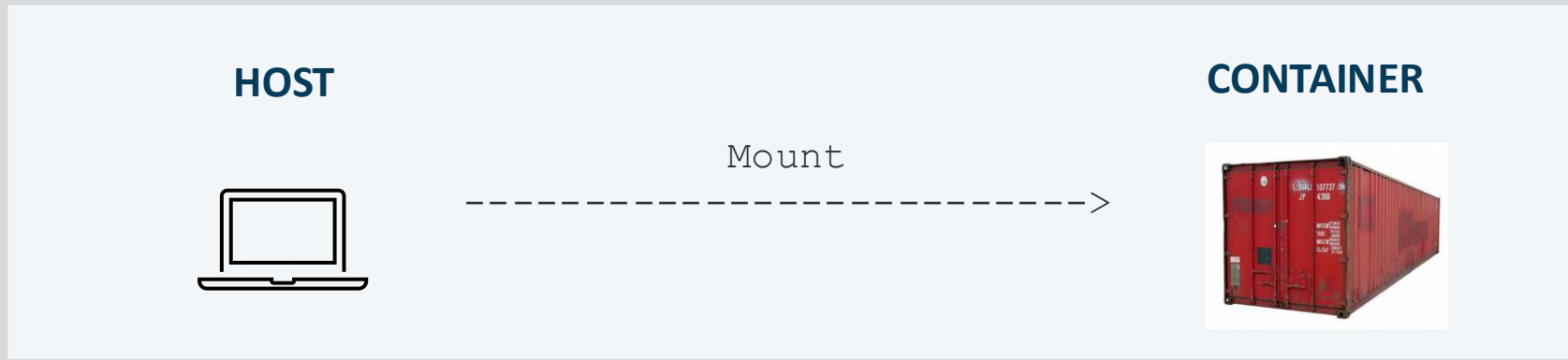
- The image may include:
 - Ubuntu
 - Python
 - pytest
 - Required packages
 - Course tools ...

You do not need to rebuild the image every time you modify your Python code!



Image Build vs. Mount

- A **mount** connects the course repository on the HOST to the container.



- When you save a file in VS Code, the updated file is **immediately** visible inside the container through `/workspace`.

4. HOST vs. CONTAINER

HOST

Your computer

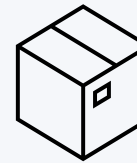
Use HOST to open VS Code, edit files, and start the Docker container.



CONTAINER

Docker lab environment

This is where Python, pytest, and lab commands should run.



5. Where Do I Edit My Code?

- Docker is the execution environment, not the storage location.
- Edit locally and run inside Docker

Do this

- Open the lab file in VS Code Explorer.
- Edit the Python code.
- Save the file.

Avoid this

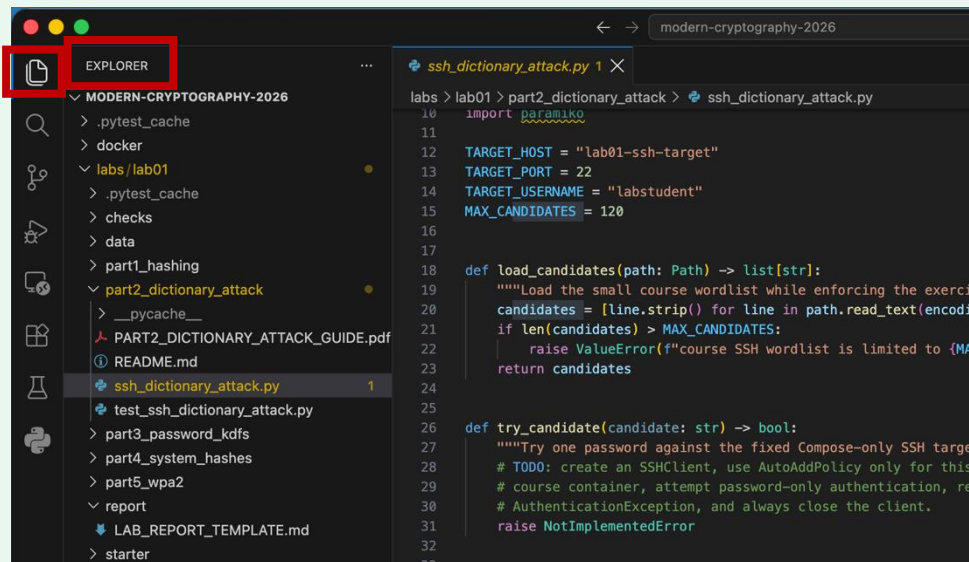
- Edit files only inside the container.
- Treat the temporary container as permanent storage



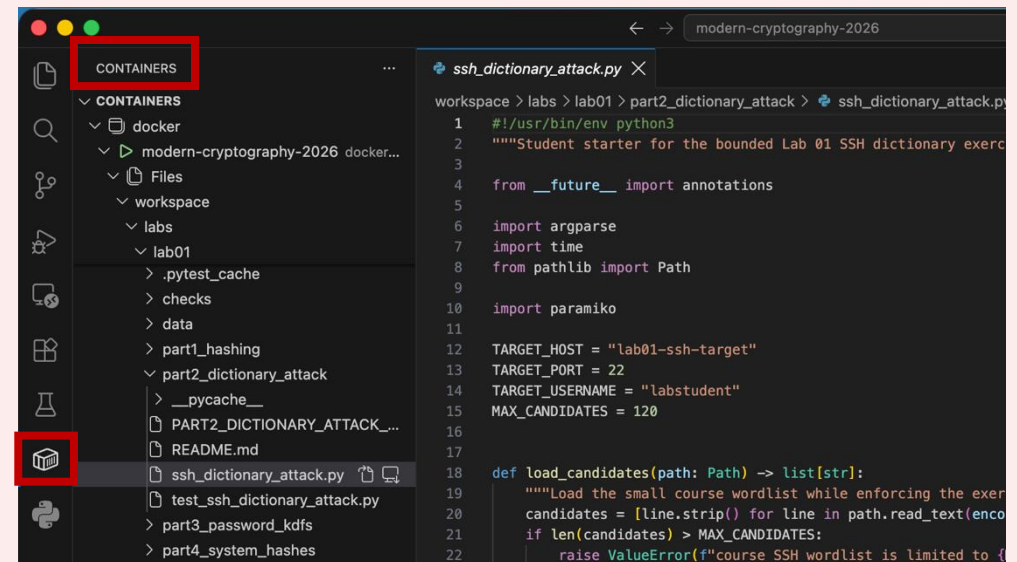
5. Where Do I Edit My Code?

- Because files inside the temporary container may disappear when the container is removed.

Edit the file in *EXPLORE*



Do not edit the file in *CONTAINERS*



6. Where Do I Run My Code?

- Run lab commands inside the Docker container.

1) Use this for automated checks.

```
cd /workspace/labs/lab01  
pytest -q
```

2) Use this for running one program.

```
cd /workspace/labs/lab01/part1_hashing  
python3 hash_password.py
```

If you run lab commands in PowerShell or the wrong terminal, the required tools may not be available.

